

Q4.

$$g = \nabla \hat{R}(w) = \frac{1}{n} \sum_{i=1}^n g_i$$

$$\sigma^2 = \frac{1}{n} \sum_{i=1}^n \|g_i - g\|^2$$

4.1

$\{1, \dots, n\}$

For an i.i.d  $X$ ,

$$E[X] = \sum_i P(x_i) \cdot x_i$$

For,  $E[g_i]$

$$E[g_i] = \sum_{i=1}^n P(f=i) \cdot g_i$$

For an i.i.d,  $P(f=i) = \frac{1}{n}$

$$E[g_i] = \sum_{i=1}^n \frac{1}{n} \cdot g_i = \frac{1}{n} \sum_{i=1}^n g_i = \nabla \hat{R}(w) = g$$

$$\text{Similarly, } E[\|g_i - g\|^2] = \sum_{i=1}^n P(Z=i) \|g_i - g\|^2$$

$$= \frac{1}{n} \sum_{i=1}^n \|g_i - g\|^2 = \sigma^2$$

4.2

B indices i.i.d  $I_1, \dots, I_B$ 

$$\hat{g}_B = \frac{1}{B} \sum_{b=1}^B g_{I_b}$$

$$E[\hat{g}_B] = \frac{1}{B} \sum_{b=1}^B E[g_{I_b}]$$

$$= \frac{1}{B} B g = g$$

$$\Rightarrow \hat{g}_B - g$$

$$= \frac{1}{B} \sum_b g_{I_b} - \frac{1}{B} \sum_b g$$

$$\hat{g}_B - g = \frac{1}{B} \sum_b (g_{I_b} - g)$$

$$\|\hat{g}_B - g\|^2 = \frac{1}{B^2} \sum_{b=1}^B \sum_{c=1}^B (g_{I_b} - g)^T (g_{I_c} - g)$$

$$E[\|\hat{g}_B - g\|^2] = \frac{1}{B^2} \sum_{b=1}^B \sum_{c=1}^B E[(g_{I_b} - g)^T (g_{I_c} - g)]$$

When  $b=c$

$$E[\|g_{I_b} - g\|^2] = \sigma^2$$

When  $b \neq c$

$$E[(g_{I_b} - g)^T (g_{I_c} - g)] = E[(g_{I_b} - g)(g_{I_c} - g)^T] = 0$$

$\Rightarrow b=c$ , for  $B$  terms

$$E[\|\hat{g}_B - g\|^2] = \frac{1}{B^2} \cdot B \sigma^2 = \frac{\sigma^2}{B}$$